

JEE MAIN REVISION MASTER: MATRICES AND DETERMINANTS SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 368 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Matrices And Determinants - Core Application	368 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Matrices And Determinants - Core Application (Very High Yield — 368 PYQs)

Core Principle: Key problem pattern in Matrices And Determinants.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "If the system of linear equations : $x+y+z=6$ $x+2y+5z=10$ $2x+3y+\lambda z=\mu$ has infinitely many s..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "If $f:N \rightarrow Z$ is defined by $f(n)=\begin{aligned} n &\& -1 &\& -5 \\ &\& -2n^2 &\& 3(2k+1) &\& 2k+1 \\ &\& -3n^3 &\& 3k(2k+1) &\& 3k(k+2)+1 \end{aligned}$..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "If the system of equations : $x+y+z=5$ $x+2y+3z=9$ $x+3y+\lambda z=\mu$ has infinitely many solutions,..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Let $A = \begin{bmatrix} 1 & 1 & -1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ satisfy $A^2 + \alpha(\text{adj}(\text{adj}(A))) + \beta(\text{adj}(A)(\text{adj}(\text{adj}(A)))) = \begin{bmatrix} 2 & -2 & 2 & -2 & 0 & -1 & 0 & 0 & -1 & 0 \\ 0 & 2 & -2 & 2 & 0 & -1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & -2 & 2 & 0 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 2 & -2 & 2 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -2 & 2 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & -2 & 2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$ for some $\alpha, \beta \in \mathbb{R}$. Then $(\alpha - \beta)^2$ is equal to ____

Correct Answer: 4

Solution Analysis: We need to find α and β from $A^2 + \alpha(\text{adj}(\text{adj}(A))) + \beta(\text{adj}(A)(\text{adj}(\text{adj}(A)))) = \begin{bmatrix} 2 & -2 & 2 & -2 & 0 & -1 & 0 & 0 & -1 & 0 \\ 0 & 2 & -2 & 2 & 0 & -1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & -2 & 2 & 0 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 2 & -2 & 2 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -2 & 2 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & -2 & 2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$.

Let us solve step by step. 1. Given matrix $A = \begin{bmatrix} 1 & 1 & -1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ 2. Find $\det(A)$ Since the third row is $(0, 0, 1)$, expand along the third row: $\det(A) = 1 \cdot \begin{vmatrix} 1 & 1 & -1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{vmatrix} = (-1)(0) - (1)(1) = -1$. So, $\det(A) = -1 \neq 0$. Hence A is invertible. 3. Use NCERT property of adjoint of adjoint For an $n \times n$ invertible matrix, $\text{adj}(\text{adj}(A)) = \det(A)^{n-2} A$.